

CONSULT 360 AI

Pre-Consultation Clinical Intelligence & Hospital Information System (HIS)

Author & Engineering: Full-Stack Healthcare Product Architecture Team
Institution: St. Jude Medical Center · Clinical Informatics Division
AI Engine: Google Gemini 3.6 Flash (Multimodal Vision OCR + Structured Synthesis)
Architecture: 3-Tier REST Micro-Architecture · 10 Database Collections

1. Executive Summary & Problem Statement

In modern hospital outpatient workflows, physicians are allocated only **10 to 15 minutes per consultation**. Patients frequently present with **years of fragmented, multi-source records**—including paper prescriptions, PDF lab panels, 12-lead ECGs, 2D echocardiograms, and chest radiographs. Manually cross-referencing these records causes physician burnout, overlooked biomarker deterioration, and missed care gaps.

Consult 360 AI ingests multi-format records, applies multimodal computer vision OCR, extracts structured clinical entities, computes longitudinal biomarker trends (Delta), detects care gaps based on ADA, KDIGO, and ACC/AHA guidelines, and delivers an **explainable 60-second Clinical Brief** before the consultation.

2. Technology Stack & Implementation Details

Layer	Technology / Framework	Implementation Details
Frontend Core	Vanilla JavaScript (ES6+)	Sub-50ms render latency without Virtual DOM overhead. Uses async/await, fetch with retries, EventSource for SSE streaming.
Design System	Custom Clinical CSS3	Epic/Cerner EHR style. Deep Teal (#0F4C5C), neutral background (#f8fafc), 8-10px corner radius, subtle borders.
Typography	Inter & JetBrains Mono	Inter for readability; JetBrains Mono for biometrics, ICD-10 codes, MRNs, dates, and delta shifts.
Client PDF	Mozilla PDF.js 3.11.174	Client-side web worker for instant PDF rasterization and text layer parsing.
Backend	Node.js & Express 4.18.2	High-concurrency runtime with modular REST controllers, CORS, and JSON body parsers.
AI Engine	Google Gemini 3.6 Flash	@google/generative-ai executing multimodal vision OCR (ECGs, Echo, X-rays) & structured JSON synthesis.
File Buffer	Multer 1.4.5-lts.1	In-memory storage buffers (25MB limit) preventing disk I/O bottlenecks.
Security	Node Crypto (PBKDF2)	SHA-256 PBKDF2 password hashing & HMAC session tokens. Zero native binary dependencies.

3. Hospital Database Architecture (10 Collections)

Collection	Records	Model Fields & Description
1. Doctors	10 Doctors	doctorId, name, email, password (PBKDF2), department, designation, experience, room, licenseNo
2. Patients	100 Records	id, mrn (MRN-2024-xxxx), name, age, gender, bloodGroup, emergencyContact, chronicConditions, allergies, riskScore, careJourney
3. Appointments	350 Records	appointmentId, patientId, doctorId, appointmentDate, appointmentTime, status (in-progress, scheduled, completed), reason
4. MedicalReports	250 Records	reportId, patientId, reportType (ECG, Echo, Labs, X-Ray), uploadedFile, OCRText, uploadedDate, verifiedBy
5. Investigations	50+ Orders	investigationId, patientId, investigationName, status (attention, pending), urgency, dueDate, guidelineRef
6. FollowUps	40+ Recalls	followUpId, patientId, followUpDate, status (overdue, scheduled), assignedDoctorId, reason, contactStatus
7. AISummaries	100 Briefs	patientId, summary, risk, confidence, entities (diagnoses, meds, vitals, labs), clinicalChanges (Delta), missingInvestigations
8. Notifications	30+ Alerts	notificationId, doctorId, title, message, priority (critical, high), type, isRead, createdAt, mrn
9. Departments	10 Wings	departmentId, departmentName, floor, headDoctor, activeDoctorsCount, totalPatients
10. AuditLogs	30+ Logs	auditId, doctorId, patientId, action (CHART_VIEW, ORDER_LAB, LOGIN), timestamp, ipAddress, details

4. Doctor-Specific Specialist Cohorts (100 Patients)

Doctor ID & Name	Specialty Department	Assigned Patients & Conditions (10 Each)
DOC1001 Dr. Amit Sharma	General Medicine	Rajesh Kumar (T2DM+HTN+Proteinuria), Ramesh Gupta (Metabolic Syndrome), Pooja Joshi (Anemia), Dinesh Deshmukh (PUO), Harold Finch (Polypharmacy), Seema Saxena, James Wilson, Barbara Taylor, Matthew Thomas, Amanda Martin.
DOC1002 Dr. Sarah Chen, MD	Cardiology	Priya Sharma (Post-viral Myopericarditis), Arjun Mehta (CAD Post-CABG), Vikram Reddy (HFrEF 35%), Sneha Chopra (AFib RVR), Steven Anderson (Hypercholesterolemia CAC 480), Ajay Mishra, Kishore Bhat, Robert Martinez, Nancy Davis, Christopher Gonzalez.
DOC1003 Dr. Rajesh Verma	Endocrinology & Diabetology	Meera Iyer (T1DM on CGM), Manoj Pillai (Diabetic Foot Ulcer), Alok Bose (Uncontrolled T2DM HbA1c 10.4%), Jyoti Menon (Hashimoto Thyroiditis), David Johnson (Cushing), Patricia Rodriguez, Daniel Hernandez, Kimberly Lopez, Anthony Clark, George Hall.
DOC1004 Dr. Ananya Iyer	Neurology & Stroke Center	Suresh Rao (Acute Stroke Post-tPA), Kavita Nair (Migraine), Harish Joshi (Parkinson), Swati Chopra (Epilepsy), Gaurav Bhat (MS), Shilpa Menon (ALS), Rahul Deshmukh, Aarti Saxena, Kiran Pillai, Pallavi Bose.
DOC1005 Dr. Michael Scott	Medical Oncology	Vikas Kumar (Colorectal Ca Stage IIb), Rashmi Sharma (Breast Ca), Pradeep Mehta (NSCLC EGFR+), Renu Patel (DLBCL), Ashok Singh (Prostate Ca), Manju Iyer, Anita Reddy, Usha Gupta, Mukesh Verma, Radha Nair.
DOC1006 Dr. Priya Nair	Nephrology & Renal Care	Sunita Singh (CKD Stage 3b), Vikram Iyer (ESRD on HD), Deepa Reddy (Membranous Nephropathy), Amit Gupta (ADPKD), Kavita Verma (Diabetic Glomerulosclerosis), Ramesh Nair, Pooja Joshi, Manoj Chopra, Sneha Rao, Naveen Bose.
DOC1007 Dr. Vikram Seth	Pulmonology & Respiratory	Divya Mishra (COPD Gold Group D), Sanjay Deshmukh (Eosinophilic Asthma), Geeta Saxena (IPF), Alok Pillai (OSA AHI 42), Swati Menon (Sarcoidosis), Harish Bhat, Neha Kumar, Gaurav Sharma, Shilpa Mehta, Rahul Patel.
DOC1008 Dr. Emily Watson	Gastroenterology	Aarti Singh (NASH F3 Fibrosis), Kiran Iyer (Cirrhosis Ascites), Pallavi Reddy (Crohn), Vikas Gupta (Ulcerative Colitis), Rashmi Verma (Pancreatitis), Pradeep Nair, Jyoti Joshi, Ajay Chopra, Renu Rao, Ashok Bose.
DOC1009 Dr. David Miller	Orthopedics & Joint Care	Seema Mishra (Knee OA Grade 4), Santosh Deshmukh (Spondylolisthesis), Manju Saxena (ACL Tear), Kishore Pillai (Rotator Cuff), Anita Menon (Osteoporosis Fracture), Dinesh Bhat, Usha Kumar, Mukesh Sharma, Radha Mehta, Vijay Patel.
DOC1010 Dr. Sunita Rao	Pediatrics & Adolescent Care	Shashi Singh (Pediatric T1DM), Gopal Iyer (Pediatric Asthma), Lata Reddy (Failure to Thrive), John Gupta (Rheumatic Fever), Mary Verma (Minimal Change Nephrotic), David Nair, Sarah Joshi, James Chopra, Patricia Rao, Robert Bose.

5. Complete List of All System Features

- **Hospital Authentication Entry Gateway:** Opens by default on URL load with quick one-click selector chips (DOC1001, DOC1002, DOC1003, password: consult360), PBKDF2 password verification, and 3-step animated verification sequence.
- **Doctor Scope Switching ([My Assigned] vs [All Hospital]):** Allows clinicians to toggle seamlessly between their 10 assigned patients and the full 100-patient hospital-wide census with dynamic KPI recalculations.
- **6-Stage Care Journey Continuum Tracker:** Visual stepper tracking Consultation -> Diagnosis -> Treatment -> Investigation -> Follow-up -> Review with completed, attention, missed, and pending status flags.
- **Longitudinal Biometric Delta (Delta) Shift Engine:** Calculates baseline vs intermediate vs current biomarker shifts (e.g. HbA1c 8.4% -> 10.4%, eGFR 88 -> 48 mL/min) with direction indicators (Worsening/Improving/Stable).

- **Multimodal Vision OCR Pipeline:** Computer vision extraction of 12-lead ECG waveforms, 2D echocardiograms, chest radiographs, and handwritten doctor notes directly from image and PDF buffers.
- **7-Stage Server-Sent Events (SSE) AI Streaming:** Real-time step-by-step progress telemetry streamed over HTTP SSE during multi-document ingestion and LLM clinical synthesis.
- **Missing Investigations & Guideline Radar:** Audits active diagnoses against international standards (ADA, KDIGO, ACC/AHA, GOLD) and flags overdue surveillance tests with a one-click Order Lab integration.
- **Physician Next-Action Directives:** Categorizes actionable guidance into Immediate Clinical Actions, Medication Reconciliation, and Patient Counseling.
- **Interactive Doctor Alert Center ([Alerts]):** Slide-out drawer with real-time notification cards for high-risk triage admissions and overdue diagnostic tests with mark-as-read controls.
- **Clinical Hotkeys & Accessibility:** Keyboard navigation (? for shortcuts modal, D for Dashboard, U for Upload, N for Notifications, F for Search, ESC to dismiss).

6. Comprehensive REST API Reference (17 Endpoints)

HTTP	Route Endpoint	Parameters / Body	Response Description
POST	/api/login	{ doctorId, password }	Authenticates doctor; returns PBKDF2 validated token & profile.
GET	/api/me	Bearer Token	Returns current authenticated doctor profile.
POST	/api/logout	Bearer Token	Clears active clinician session.
GET	/api/doctors	None	Returns list of all 10 hospital specialist doctors.
GET	/api/patients	?doctorId=DOC1001&search;=...	Returns filtered patient records with pagination.
GET	/api/patient/:id	:id (e.g. PAT1001)	Returns full patient chart, reports, and AI brief.
POST	/api/patient	{ name, age, condition, ... }	Registers a new outpatient in hospital database.
GET	/api/dashboard	?doctorId=DOC1001	Returns real-time computed hospital KPIs & worklist.
GET	/api/appointments	?doctorId=DOC1001	Returns scheduled appointments ledger.
GET	/api/investigations	?status=attention	Returns active & pending diagnostic lab orders.
POST	/api/investigations/order	{ patientId, testName }	Clinician places a new diagnostic order.
GET	/api/followups	?status=overdue	Returns outpatient recall schedule records.
GET	/api/notifications	?unreadOnly=true	Returns real-time alerts for the active doctor.
POST	/api/notifications/:id/read	:id	Marks doctor notification alert as read.
POST	/api/upload-and-analyze	multipart/form-data	7-Stage SSE streaming pipeline for upload & AI synthesis.
POST	/api/summary	{ text, patientId }	Generates structured Gemini 3.6 Flash clinical brief.
GET	/api/health	None	Health check reporting version, database counts, and Gemini status.

7. Summary & Clinical Impact

Consult 360 AI transforms outpatient healthcare by bridging the gap between massive, unstructured medical records and high-pressure 15-minute consultations. By pairing **Google Gemini 3.6 Flash multimodal vision** with a robust **Node.js REST architecture**, a **10-collection Hospital Database**, and an **EHR-grade clinical design system**, the platform delivers explainable, guideline-grounded pre-consultation intelligence that enhances diagnostic accuracy, prevents overlooked care gaps, and drastically reduces physician administrative burnout.

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